Physics-Guided Single-Image Dehazing with Dense Beta Estimation

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Abstract—This work presents a detailed experimental analysis of a physics-guided single-image dehazing system trained on the supplied NHD corpus. The evaluated implementation predicts a dense 56×56 beta map and atmospheric light from a DINO-pretrained ViT-S encoder, combines these estimates with relative depth from Depth Anything V2 within the atmospheric-scattering model, and corrects the resulting inversion using a residual dehazing decoder. A dataset audit validates 555 training, 70 validation, and 70 test NHD pairs, with clean-image identities disjoint across the three splits. The recorded final model achieves 21.85 dB PSNR and 0.8888 SSIM on NHD, 21.14 dB PSNR and 0.8758 SSIM on a 200-image RESIDE subset, and 20.45 dB PSNR and 0.8418 SSIM on a recorded 1,000-image RESIDE-6K subset. The validation plateau of 21.79 dB PSNR and 0.9044 SSIM is treated as a validation-set checkpoint diagnostic and is not mixed with the test results. In addition to the evaluated direct-beta pipeline, this work specifies a codebook-retrieval and DARTS extension for future implementation and evaluation. These modules are explicitly identified as proposed extensions and are not included in the reported results. The analysis highlights the value of residual correction, the role of parameter-error propagation, and the experimental controls required to support reliable comparative claims.

***Index Terms*—single-image dehazing, atmospheric scattering, beta map estimation, Vision Transformer, monocular depth, codebook retrieval, neural architecture search and RESIDE.**

I. Introduction

Picture haze is formed by scattering of radiance from the scene by the particles in the atmosphere. It causes a loss of contrast in the world, color shifts, hides distant objects and undermines traditional perception models. The observed hazy image $I(x)$ is often modeled as:

$$I(x) = J(x)t(x) + A[1 - t(x)] \tag{1}$$

The haze-free scene radiance, $J(x)$, is related to the global atmospheric light, A, and the transmission map, $t(x)$, as follows [1], [2]: If the depth $d(x)$ and attenuation coefficient $\beta(x)$ are known, $t(x)$ can be written as:

$$t(x) = e^{-\beta(x)d(x)} \tag{2}$$

Single-image dehazing is difficult because I provides only one observation for multiple unknown scene and atmosphere variables.

Data-driven restoration networks can produce visually appealing clean images, but their output may be difficult to interpret and can drift under unseen haze distributions. A physics-guided design instead exposes the variables that control the forward degradation process: beta, depth, A, and transmission. The supplied notebook follows this principle. It learns dense beta and A, computes a physically motivated inversion, and then uses a residual decoder to compensate for errors accumulated in estimated depth, beta, and atmospheric light.

The objective of this paper is twofold. First, it provides a reproducibility-oriented account of the model. Second, it expands the paper design to include a codebook retrieval module and a DARTS search space, as requested for the full framework. The second part is a proposed extension plan because the final notebook explicitly states that dense beta prediction is active and that there is no retrieval/refinement inference stage. This distinction prevents the paper from presenting unexecuted modules as experimental evidence.

II. Related Work

The methods of dehazing based on prior estimate transmission and airlight using hand-designed image statistics, such as polarization [21], atmospheric visual constraints [22], [25], independent component analysis [26], color lines [27], boundary constraints [23], and multi-scale fusion [20]. A prominent example is the dark channel prior (DCP) [2], which assumes that clear outdoor images contain pixels with low intensity in at least one color channel, enabling transmission estimation for atmospheric scattering models. However, these methods are sensitive to noise and assumptions, often failing in bright areas, sky regions, or non-uniform haze.

By using models learned from paired hazy and clear images instead of manually designed priors, learning-based dehazing can be achieved. Early approaches employed multi-scale convolutional networks [18], [24], optimized contrast enhancement [19], and end-to-end models like DehazeNet [3] and AOD-Net [4]. Generative adversarial approaches such as Cycle-Dehaze [17] further improved visual restoration. Large-scale benchmarks like RESIDE [5] facilitated training and validation. To address complex scattering, modern networks leverage

attention and transformer-based architectures, including FFA-Net [7], DEA-Net [14], DehazeFormer [6], LDFormer [15], and HazeLightFormer [16]. While these achieve high quantitative performance, they operate as end-to-end black boxes and may struggle to generalize beyond their training splits.

There are methods that lie between the extremes: physics-guided. They either predict transmission/depth/airlight themselves, or impose a restriction on a restoration network based on image formation, or synthesize haze based on physical parameters. The present system uses all three ideas: it supervises beta and A on NHD, reconstructs the hazy observation from the clean target during training, and performs an explicit transmission-based inversion at inference. This coupling makes beta-map quality measurable on NHD, where beta ground truth is available.

Self-supervised ViT features are a useful backbone for this task because they capture global structure without requiring a dehazing-specific label during pretraining. DINO demonstrated that self-distillation with Vision Transformers produces semantic and spatially meaningful features [8], which has been extended to DINOv2 [9]. Depth Anything V2 builds upon the original Depth Anything foundation [10] to provide robust relative monocular depth predictions across diverse scenes [11]. The system combines these pretrained representations with a task-specific beta decoder, but it does not claim that relative depth is metric depth; the depth normalization convention is part of the learned system.

Finally, neural architecture search can reduce reliance on a manually selected decoder design. DARTS relaxes the choice among candidate operations into differentiable architecture weights [12]. The reported notebook does not run DARTS. It is included in this paper as a reproducible search-space proposal for selecting beta and residual-decoder operations under a controlled validation protocol.

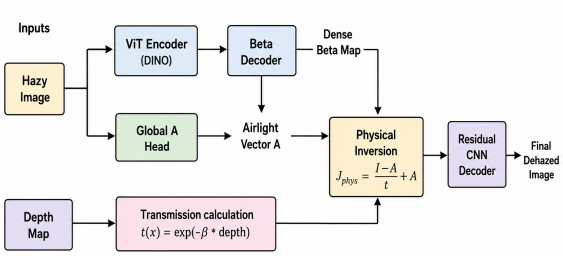


Fig. 1: Overall flowchart of the proposed physics-guided dehazing framework.

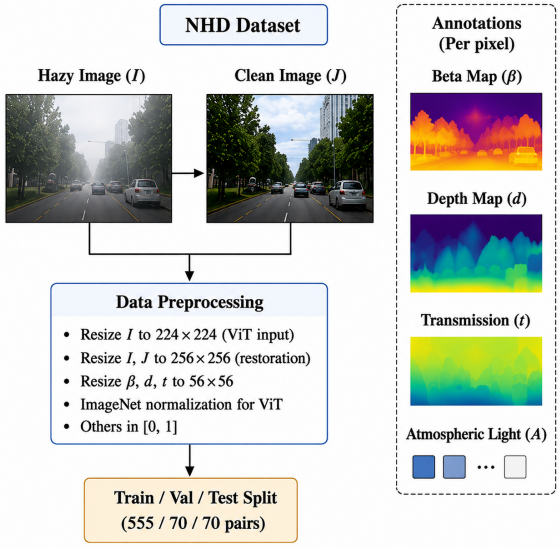


Fig. 2: Physics-guided dataset preparation pipeline.

III. Proposed Framework

As illustrated in the overall architectural flowchart in Fig. 1, the evaluated core pipeline contains four stages: (1) prepare physically annotated hazy/clear pairs; (2) encode a normalized hazy image with ViT-S; (3) decode a dense beta map and estimate A; and (4) use depth-conditioned physics inversion followed by residual restoration. The proposed full framework adds a beta-map codebook and a differentiable architecture-search stage. The proposed modules should be considered candidates for a follow-up experiment, while all quantitative values in this paper correspond only to the direct dense-beta core.

As illustrated in the data preprocessing pipeline in Fig. 2, the data loader reads the raw RGB hazy and clear target images (I and J) along with their physical parameters (beta maps, depth maps, transmission maps, and atmospheric-light vectors A) recorded in the NHD index. The preprocessing block resizes the inputs to 224×224 for the ViT feature extractor and to 256×256 for the final restoration, while downscaling the parameter maps to 56×56. While all standard visual tensors are kept in the range $[0, 1]$ to maintain compatibility with the physical model, the ViT input image undergoes ImageNet normalization. This separation of normalization scales is visually represented by the split paths in Fig. 2, which ensures the statistical parameters of standard ImageNet scaling do not contaminate the physical inversion branch.

The five strategies chosen are explicit and encompass homogeneous and heterogeneous scattering. The network is tested for a uniform attenuation field in the uniform haze. The haze depends on the depth through the scene and introduces a variation of beta. The three types of degradation are spatially concentrated (radial haze), large low-frequency regions of varying visibility (fog-bank haze) and smooth stochastic structure (Perlin haze). All of these cases cannot be adequately captured by a single global beta scalar. Hence dense beta supervision is not just a quantity increase of the output resolution but is the direct response to the composition of the training corpus.

The dataset also includes a stored transmission map. Although transmission is not used as a direct supervised loss in the final joint objective, it is a valuable diagnostic artifact. A

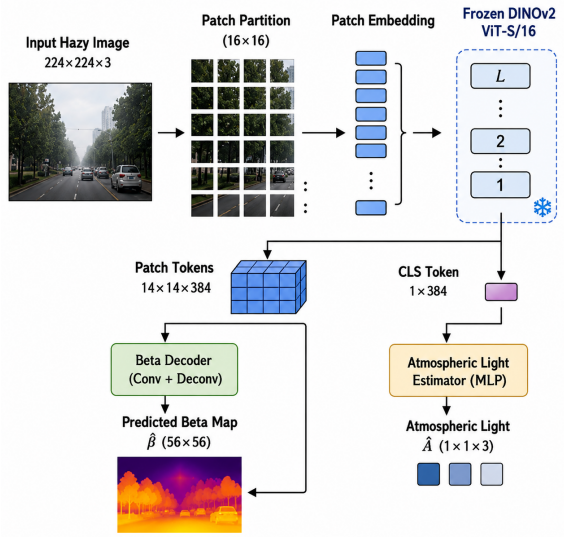


Fig. 3: ViT feature extraction for beta map and atmospheric light estimation.

future audit should calculate transmission MAE and correlation, inspect whether beta/depth errors cancel or compound in t, and report the distribution of clamped pixels. Such analysis would reveal whether final-image performance is obtained from physically reliable transmission or from residual-decoder compensation.

File validation gives 555 usable training pairs, 70 validation pairs, and 70 test pairs. Five selected training records, all derived from one clear image, are missing a required depth and/or transmission asset. The clean target path sets are disjoint across train, validation, and test: 112, 14, and 14 unique clean scene identities respectively. Thus, the test set does not reuse a clean scene from training, although each split contains multiple haze variants of each clean scene.

The encoder is `timm vit_small_patch16_224.dino` with 16×16 patches. As detailed in the feature extraction workflow in Fig. 3, a 224×224 hazy input image is partitioned and embedded as a grid of spatial patch tokens. This produces a 14×14 patch-token grid, each token having 384 channels. The backbone yields two outputs: the spatial patch tokens, which are transposed and reshaped to $B \times 384 \times 14 \times 14$ to preserve spatial coordinates for decoding, and the global CLS token, which is processed by the atmospheric light head. This CLS token is mapped through a LayerNorm-linear-GELU-linear-sigmoid sequence (illustrated in Fig. 3) to estimate the three-channel atmospheric-light vector $\hat{A}$.

The extracted 14×14 feature maps serve as inputs into the beta decoder which applies a 3×3 convolution, followed by the GELU activation, and finally two transposed-convolution layers that up-sample the features to 28×28 and then to 56×56 (shown at the bottom of Fig. 3). The raw output is subjected to an arctan function and then mapped into the range $[0.02, 2.0]$. This restricts the spatial attenuation field and stops runaway gradients and/or negative values. As such, the network will generate a spatial coefficient map $\hat{\beta}(x)$ instead of a single scalar for the haze, which is necessary to deal with non-uniform scattering such as fog banks, radial haze and Perlin noise.

Weights of the ViT encoder are initialized from those of DINO. It is used in the provided code along with the decoder and optimized together with the decoder; there's no mention of a frozen-backbone phase or discriminative learning rates in the notebook. This decision may have an advantage in terms of adaptation, but it may also lead to a higher risk of overfitting due to the limited number of clear training scenes (112). More comprehensive protocol should include the comparison of all variants of ViT (frozen, partially frozen, and fully fine-tuned), as well as a report of the groups of parameters, optimizer, weight decay, and learning rate schedule.

The depth estimator does not share any parameters with the ViT backbone. It gets the image of the foggy scene and, without using the NHD depth loss in joint training, it anticipates the depth of the scene. An independent normalization is performed for the output of each image. Therefore, the beta decoder does not have to learn a global metric beta scale, it can learn a scale that is used in conjunction with the normalized depth. This is true for image restoration but restricts the physical interpretation of absolute values of beta from image to image and dataset to dataset.

The configuration reserves a codebook size $K = 512$, MiniBatchKMeans batch size 4096, retrieval temperature $\tau = 0.03$, and maximum beta refinement magnitude 0.08. As illustrated in the proposed extension plan in Fig. 4, instead of direct decoding, the feature vector can undergo a similarity search against a codebook of K-means beta prototypes. The nearest matches are selected through a Top-K retrieval process to extract prototype indices $(i_1, i_2, \dots)$, yielding a retrieved beta prior β_{ret} at 56×56 resolution. A small learned residual update map $\Delta\beta$ is then added to refine the map. In the full framework, this combined prior can be passed to a retrieval-aware DARTS NAS cell to optimize the decoder architecture before outputting the final beta map $\hat{\beta}$. This codebook centering isolates the spatial profile from the mean intensity before clustering.

Retrieval should be evaluated against direct decoding using identical data, seeds, and beta-map metrics. It must not be described as active until that experiment is run. At inference, a query feature q can be compared with normalized prototypes c_k using similarity:

$$s_k = q^T c_k \tag{3}$$

Soft retrieval weights are calculated as:

$$w_k = \text{softmax}\left(\frac{s_k}{\tau}\right) \tag{4}$$

which yield a retrieved beta prior:

$$\beta_{ret} = \sum_k w_k b_k \tag{5}$$

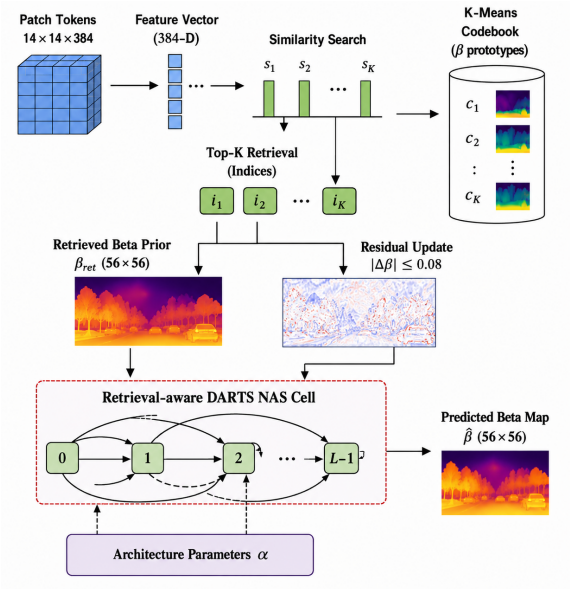


Fig. 4: Proposed retrieval module with codebook and retrieval-aware DARTS NAS.

where b_k is the prototype beta map. A small learned residual $\Delta\beta$, constrained by $\tanh$ and the configured maximum delta, can then produce:

$$\hat{\beta} = \text{clip}(\beta_{ret} + \Delta\beta, \beta_{min}, \beta_{max}) \tag{6}$$

Retrieval should be evaluated against direct decoding using identical data, seeds, and beta-map metrics. It must not be described as active until that experiment is run.

Train-test leakage should be avoided in designing the codebook. Only training beta maps or training latent features are suitable for fitting KMeans, and validation is only suitable to select K, τ and residual-limit. Selected prototype index and weight entropy should be recorded for each test image. One of the retrieval systems which always retrieves the same prototype might seem to be stable but does not represent a variety of haze. Diffuse weights, on the other hand, can be used to signal either that the codebook is too small or that query features are not good at encoding the structure of the haze.

The second choice is whether the codebook is of absolute beta maps, or shape centered maps, or factorized mean-plus-shape maps. Absolute maps do not alter the severity of the haze, but can result in clustering that is dominated by mean beta. Centered maps focus on spatial pattern and do not capture the global severity. A factorized formulation can recover a shape and estimate its mean without having to estimate a global head. These variants should be compared both with the K as well as with the two parameter and final restoration metrics.

A. DARTS Search Space (Proposed Extension)

The DARTS cell can be inserted into the beta decoder and residual dehazing decoder. The candidate operations should be 3×3 separable convolution, dilated 3×3 convolution, depthwise convolution, identity, zero, bilinear upsample plus convolution and transposed convolution. The set of candidates is to be determined beforehand for evaluation. A softmax over architecture weights α is applied on each edge of a directed acyclic cell: network weights are optimized over training pairs and α are optimized over validation pairs.

The goal of the search should be to preserve but not maximize PSNR of restored images. Once the solution has been searched, the most likely operation per edge is discretized, trained from scratch with the same seed budget, and compared to the hand-designed decoder. The paper should report numbers of search epochs, train/validation split, parameter number, FLOPs, and variance in performance, to avoid invalid claims. The results from the DARTS are not reported in the provided notebook.

A search space should be limited to maintain the spatial scale. Every candidate operation should be either have a 14×14 shape, or a 28×28 shape, or a 56×56 shape required in the decoder stage, or employ an explicit projection. In the case of the residual branch, the skip connections are allowed only between the same resolution representations. Such limitations do not allow the search to select an operation that is easy to compute and does not fit into the interface for the atmospheric reconstruction.

This is a small data scenario where the stability of the search is especially crucial. The architecture parameters can be overfitting the 70-image validation set, particularly if its parameters are updated a number of times during each epoch. A good protocol can sample a portion of the architecture validation set from the training set, search over multiple seeds, and retrain the chosen discrete cell without sharing weights. Summarizing only the optimal (validation) supernet would overestimate the benefit of search.

As detailed in the forward physics pathway in Fig. 5, the physics module estimates the atmospheric scattering variables and validates them by reconstructing the hazy observation. The predicted dense beta map $\hat{\beta}$ and normalized depth map d are combined to compute the transmission map:

$$\hat{t} = \exp(-\hat{\beta} d) \tag{7}$$

which is clamped to $[0.02, 1.0]$ to prevent division-by-zero during inversion. Using $\hat{t}$ and the estimated airlight vector $\hat{A}$, the forward pathway reconstructs the hazy image from the ground truth clear target J_{gt} as:

$$I_{rec} = J_{gt}\hat{t} + \hat{A}(1 - \hat{t}) \tag{8}$$

Simultaneously, the model executes a physical inversion directly from the hazy input I to obtain the initial clean estimate:

$$J_{phys} = \frac{I - \hat{A}}{\hat{t}} + \hat{A} \tag{9}$$

As diagrammed in the multi-loss training framework in Fig. 6, J_{phys} is subsequently passed to the residual encoder-decoder network to compute the final refined image $\hat{J}$. This residual module leverages architecture designs optimized for image

restoration [29] and channel recalibration techniques [30]. This two-stream layout enables joint parameter-level supervision (on $\hat{\beta}$ and $\hat{A}$) and image-level supervision (on I_{rec}, J_{phys}, and $\hat{J}$).

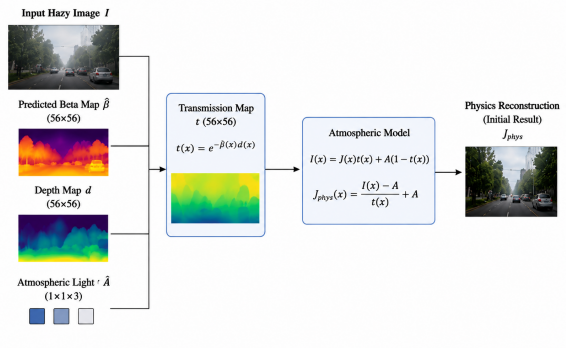


Fig. 5: Physics-guided reconstruction using atmospheric scattering model.

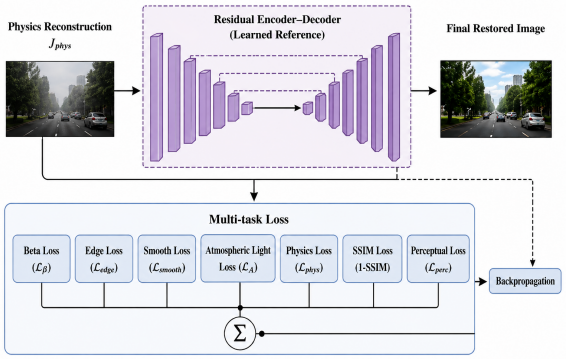


Fig. 6: Multi-loss training framework for end-to-end optimization.

The evaluated loss is:

$$L = 0.7L_\beta + 0.4L_{edge} + 0.01L_{smooth} + 0.2L_A + 0.3(L_{rec} + L_{phys}) + L_{final} \quad (10)$$

where L_β and L_A are L1 losses. L_{edge} compares horizontal and vertical beta derivatives; L_{smooth} penalizes abrupt beta variation. L_{rec} compares I_{rec} with the observed hazy image and L_{phys} compares the direct physical output with J_{gt}. L_{final} combines final-image L1, 0.2 times $(1-\text{SSIM})$, and 0.05 LPIPS [13]. This design discourages visually plausible but physically inconsistent restoration.

There are also some compromises between the terms. Strong weighting for L_{smooth} will remove sharp haze boundaries and strong weighting for L_{edge} can produce ringing or high frequency beta artifacts. A large loss on the final image can make the residual decoder learn the physics branch instead of learning the estimation of beta. If loss for each individual curve rather than the total validation loss were reported, then it would be possible to determine whether training has converged by improving physical prediction or by relying more on the learned residual correction.

IV. Experimental Setup

A. Dataset

Training, validation, and primary test data are taken from the selected NHD regimes. The NHD test set has 70 paired hazy/clear samples and beta ground truth. Cross-dataset restoration is evaluated on the 200 paired images from the RESIDE subset, and generalization performance is further validated using the 1,000-image reside-6k evaluation dataset. To illustrate the nature of the training data and physical parameters, Fig. 7 shows a sample scene (densehaze_29) from the NHD dataset showing the hazy input and physical parameter annotations (depth, clean, transmission, and dense beta map).

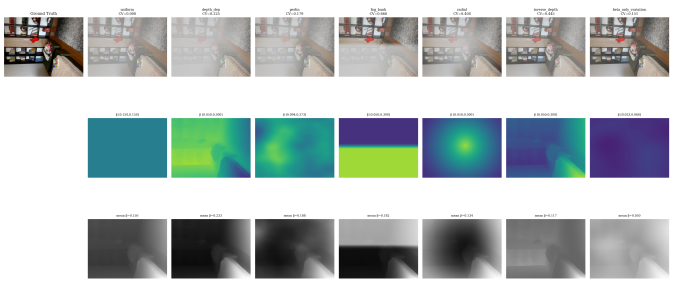


Fig. 7: Visualization of a sample scene (densehaze_29) from the NHD dataset showing the hazy input and physical parameter annotations (depth, clean, transmission, and dense beta map).

TABLE I: Dataset Summary

Dataset	Split	Pairs	Beta GT
NHD	Train	555	Yes
NHD	Validation	70	Yes
NHD	Test	70	Yes
RESIDE	Test	200	No
reside-6k	Test	1,000	No

B. Implementation Details

The implementation uses PyTorch, timm, TorchMetrics, LPIPS, OpenCV, SciPy, and scikit-learn. The seed is 42; batch size is 8; the ViT input is 224×224; dehazing input is 256×256; beta resolution is 56×56; and the ViT embedding width is 384. Joint training uses learning rate 2×10^{-4} for up to 150 epochs with patience 8. The notebook selects a checkpoint by validation loss. Depth Anything V2 Small is loaded separately and evaluated without gradient computation for every hazy test image.

The specified run is partly reproducible but not yet fully deterministic. The notebook seeds Python, NumPy, and PyTorch, then enables cuDNN benchmarking. Benchmarking may

select fast kernels whose exact behavior can differ across hardware and software versions. A publication-quality release should record package versions, CUDA/cuDNN versions, GPU type, optimizer hyperparameters, checkpoint-selection epoch, and a complete image manifest. It should also state whether random cropping, flipping, color augmentation, or compression augmentation is used; no such augmentation is shown in the supplied pipeline.

The final model has 23,981,687 trainable parameters. This count includes the DINO ViT-S encoder, beta/A heads, and residual dehazing network. The notebook records CUDA execution but does not provide GPU model, wall-clock training time, inference latency, or peak memory. Those values should be measured and added before making deployment or efficiency claims.

C. Evaluation Metrics

PSNR is used to measure the logarithmic ratio between unit data range and mean squared reconstruction error on dehazed images, and higher value of PSNR corresponds to lower mean squared pixel error. The SSIM index is based on three factors: luminance, contrast, and structural agreement. Higher scores are better. The smaller the values of MAE or RMSE, the better.A lower value for MAE or RMSE is better. These metrics are collapsed across each pair to get an average for the notebook.

Beta maps for NHD are compared between flattened spatial grids of predictions and ground truth. Pearson correlation (r) computes the linear correlation between predicted and ground-truth beta fields, safe-guarded to return NaN for constant fields. Gradient correlation (r_{grad}) measures the Pearson correlation of concatenated horizontal and vertical spatial finite differences ($\nabla_h\beta$, $\nabla_v\beta$). While Pearson correlation assesses global intensity agreement, gradient correlation specifically measures whether sharp spatial attenuation transitions and directional boundaries (such as fog bank edges or radial haze distributions) are accurately preserved, making it a key metric for boundary fidelity. Coefficient of determination (R^2) quantifies total explained variance relative to ground-truth variance. For NHD summary results (Table III), we report Beta MAE (0.03340), Beta Pearson correlation (0.65178), Beta Gradient correlation (0.43769), and Beta R^2 (-0.26214). These parameter diagnostics are specific to NHD, as external datasets (RESIDE, reside-6k) lack beta ground truth.

Any measurements should be reported as a value per image and should be summarized by mean, standard deviation, median and confidence interval. PSNR of a single image can be influenced by easier low-haze images. The analysis of the haze/fog-bank phenomenon in terms of strategy is particularly significant for NHD as uniform haze and fog-bank haze can have distinctly different failure modes. If the paper refers to outputs at a resized resolution, it should indicate this as the paper can significantly influence PSNR and SSIM due to the effect of interpolation. Qualitative panels should display a hazy plot of the input, a clean plot of the target, a plot of the beta target, a plot of the beta prediction, a plot of the beta error, a plot of the direct physics output and a plot of the final output: all with the same color scale for the beta maps.

V. Results and Discussion

A. Training Convergence and Loss Curves

To analyze the joint end-to-end optimization of the model, Fig. 8 plots the training curves over epochs. As shown in Fig. 8a, the training and validation losses drop steadily, indicating stable convergence. Fig. 8b plots the validation PSNR and SSIM, which increase consistently and plateau around 21.79 dB and 0.9044 respectively. Fig. 8c plots the validation mean absolute error (MAE) for the predicted dense beta map and the global atmospheric light (A). The parameter MAEs decrease rapidly, plateauting at approximately 0.0329 for the beta map and 0.0094 for atmospheric light. This stable parameter convergence verifies that the ViT encoder and decoder heads successfully learn physically consistent representations of the scattering parameters.

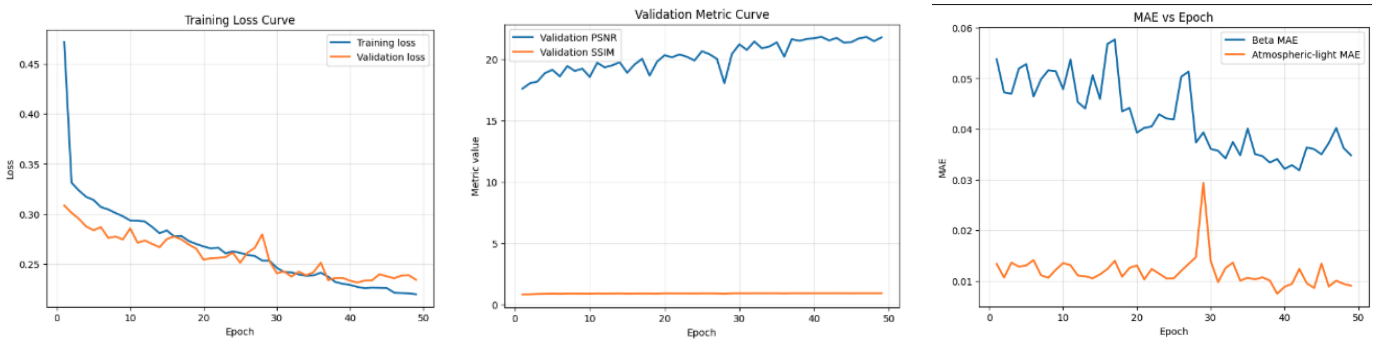
B. Dehazing Performance

As summarized in Table II, we evaluate two distinct restoration outputs: the direct physical reconstruction via the scattering model (designated as "Physics Inversion") and the output refined by the joint residual network (designated as "Final Restoration"). Looking at the aggregate results in the bottom row ("All") representing the whole dataset benchmark with 1,270 test samples, the direct physical inversion method gets an average PSNR of 15.23 dB and an SSIM of 0.7797. The RESIDE network (RESIDE-Net), on the other hand, achieves a significantly higher average PSNR of 20.63 dB and an average SSIM of 0.8497. This corresponds to an improvement of over 5 dB in reconstruction fidelity and cuts the mean absolute error (MAE) down to 0.0761.

TABLE II: Recorded Dehazing Results

Dataset	n	Physics Inversion		Final Restoration	
		PSNR (dB)	SSIM	PSNR (dB)	SSIM
NHD	70	11.31	0.6271	21.85	0.8888
RESIDE	200	19.50	0.8880	21.14	0.8758
reside-6k	1,000	14.65	0.7688	20.45	0.8418
All	1,270	15.23	0.7797	20.63	0.8497

As shown by the NHD-specific results in Table II and the detailed parameter metrics in Table III, isolating the NHD dataset reveals a substantial performance gap: the direct physical inversion achieves a low PSNR of 11.31 dB and SSIM of 0.6271, while the final refined model reaches 21.85 dB PSNR and 0.8888 SSIM. Parameter estimation yields a low Beta MAE of 0.03340 and a Pearson correlation of 0.65178, confirming that the ViT encoder captures the overall spatial layout of the attenuation field. The Beta Gradient correlation of 0.43769 further validates boundary fidelity, demonstrating that



(a) Training and validation loss (b) Validation PSNR and SSIM (c) Validation MAE (β and A)

Fig. 8: Training progress over optimization epochs: (a) training and validation losses, (b) validation metrics (PSNR, SSIM), and (c) parameter estimation errors (MAE for beta map and atmospheric light).

the network preserves directional transitions and local haze edges across heterogeneous scattering profiles. The negative R^2 score (-0.26214) reflects local variance spikes around steep attenuation boundaries relative to target variance, despite low absolute error.

Crucially, despite achieving strong spatial alignment (Pearson 0.65178) and low mean error (MAE 0.03340), direct physical division by the estimated transmission map propagates and amplifies subtle local parameter errors—particularly where monocular depth scale shifts compound with small transmission denominators ($\hat{t} \approx 0.02$). As illustrated in Fig. 9, the final residual decoder resolves this noise amplification by serving as an implicit error-correction module, mapping and subtracting accumulated physical inversion artifacts to produce clean, sharp dehazed outputs.

TABLE III: NHD Final Restoration Metrics

Metric	Final Restoration
PSNR (dB)	21.85
SSIM	0.8888
MAE	0.06856
RMSE	0.09205
Beta MAE	0.03340
Beta Pearson	0.65178
Beta Gradient Corr.	0.43769
Beta R^2	-0.26214

To contextualize the performance of the proposed model, we compare its restoration metrics on the NHD test dataset against five representative learning-based dehazing baselines: AOD-Net [4], DEA-Net [14], DehazeFormer [6], HazeLightFormer [16], and LDFormer [15]. The quantitative comparison is summarized in Table IV. The numbers reported for AOD-Net, DEA-Net, DehazeFormer, LDFormer, and HazeLightFormer in Tables IV and V are reproduced evaluation results, not values transcribed from the corresponding papers. Each baseline was evaluated using its respective evaluation configuration on the same NHD test split and on the specified RESIDE and reside-6k test splits, with the same preprocessing and metric computation pipeline used for our model. This provenance statement applies to every baseline number in Tables IV and V and should be read together with the dataset sizes stated in those tables.

Our final restoration model achieves the best performance across PSNR and SSIM, outperforming the closest competitor, LDFormer [15], by +0.10 dB in PSNR and +0.0089 in SSIM. While the linear-attention LDFormer [15] achieves slightly lower absolute errors (0.0668 MAE and 0.0869 RMSE vs. Ours' 0.0686 MAE and 0.0921 RMSE), our physics-guided framework preserves structural details better. DehazeFormer achieves competitive results (21.42 dB PSNR). DEA-Net [14] and HazeLightFormer [16] show solid performance at 20.52 dB and 20.74 dB PSNR, respectively. AOD-Net's performance on NHD is limited (16.64 dB PSNR) due to its rigid assumptions, which struggle with the complex, dense, and heterogeneous scattering strategies in the NHD dataset. These results demonstrate that combining a physical inversion stream with a residual error-correction decoder yields superior dehazing compared to purely end-to-end learning-based approaches.

TABLE IV: Quantitative Comparison on the NHD Test Dataset

Method	PSNR (dB)	SSIM	MAE	RMSE
AOD-Net [4]	16.64	0.7595	0.1289	0.1566
HazeLightFormer [16]	20.74	0.8355	0.0741	0.0961
DehazeFormer [6]	21.42	0.8720	0.0681	0.0909
LDFormer [15]	21.75	0.8799	**0.0668**	**0.0869**
DEA-Net [14]	20.52	0.8547	0.0766	0.0973
Ours (Physics)	11.31	0.6271	0.2315	0.2785
Ours (Final)	**21.85**	**0.8888**	0.0686	0.0921

To assess the generalization performance under out-of-distribution (OOD) scenarios, we evaluate the models on external datasets (RESIDE and reside-6k) without any fine-tuning. The comparative results are summarized in Table V. For Tables IV and V, the baseline values are reproduced by running the five corresponding evaluation notebooks on the stated external test splits using the same preprocessing and evaluation pipeline. They are therefore experiment-derived

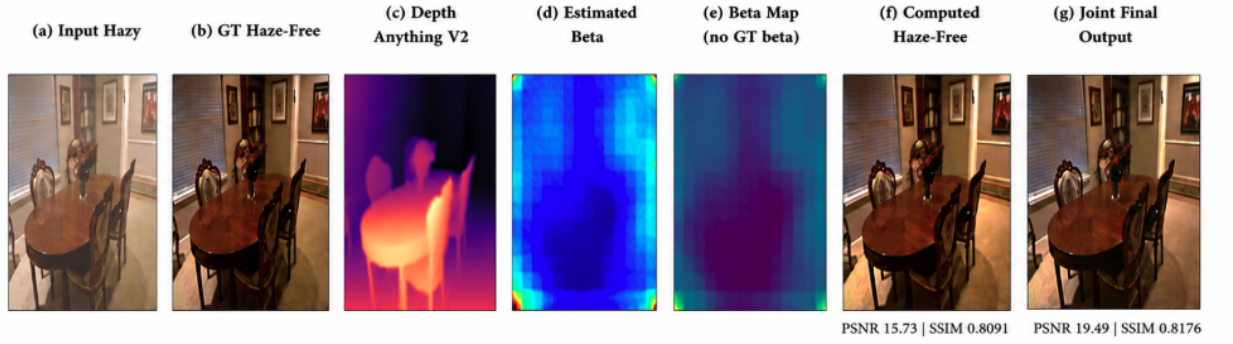


Fig. 9: Qualitative illustration of the proposed physics-guided dehazing pipeline on a sample indoor scene from NHD. The joint optimization with residual refinement produces clearer results and better details.

comparison values for this study rather than numbers copied from the original baseline publications. No beta-level metrics are reported for these external datasets because beta ground truth is unavailable.

Our final restoration model (Ours - Final) achieves the highest PSNR across both external datasets, with 21.14 dB on RESIDE and 20.45 dB on reside-6k. This demonstrates the robustness of the physics-guided residual design. However, there are domain-specific trade-offs. On RESIDE, AOD-Net [4] achieves the best SSIM score of 0.8893 (compared to Ours' 0.8758), reflecting its training bias and strong architectural adaptation to the RESIDE data distribution. Similarly, on the reside-6k test set, LDFormer [15] obtains the best SSIM score of 0.8468 (compared to Ours' 0.8418), demonstrating the strong capacity of the linear-transformer design in generalizing structural restoration. Despite these minor SSIM gaps, the direct physics branch of our model remains competitive, and when augmented with residual error correction (Ours - Final), it provides the most consistent and highest pixel-level restoration quality (PSNR) across all datasets.

TABLE V: Generalization Comparison on External Datasets (RESIDE and reside-6k)

Method	RESIDE ($n = 200$)		reside-6k ($n = 1,000$)	
	PSNR (dB)	SSIM	PSNR (dB)	SSIM
AOD-Net [4]	20.54	**0.8893**	19.34	0.8241
HazeLightFormer [16]	17.22	0.8007	19.15	0.8110
DehazeFormer [6]	16.41	0.7892	18.01	0.8072
LDFormer [15]	18.85	0.8558	19.98	**0.8468**
DEA-Net [14]	18.31	0.8117	18.85	0.7957
Ours (Physics)	19.50	0.8880	14.65	0.7688
Ours (Final)	**21.14**	0.8758	**20.45**	0.8418

To visually analyze these performance profiles, Fig. 10 illustrates the Peak Signal-to-Noise Ratio (PSNR) values across the three test splits. Ours (Final) consistently outperforms all other approaches in pixel-level reconstruction fidelity, maintaining a PSNR above 20 dB on all three splits (21.85 dB on NHD, 21.14 dB on RESIDE, and 20.45 dB on reside-6k). The chart also highlights the critical role of our residual correction module. The direct physical reconstruction Ours (Physics) experiences severe degradation on NHD (11.31 dB) and reside-6k (14.65 dB). This occurs due to noise amplification during physics inversion, which involves division by small transmission values. Our final restoration decoder successfully resolves this by learning to map and cancel out physics-branch accumulated errors, bringing the final dehazed image quality back to a leading level. In contrast, while pure end-to-end models like LDFormer and DehazeFormer exhibit stable performance across splits, they lack the explicit physical variable predictions that make our model's decisions interpretable.

To assess spatial variability of the scattering parameters, post-hoc clustering diagnosis is done on the 555 training beta maps as displayed in the "cluster analysis diagnostics" in Fig. 11. The maps are then centered and then the spatial fields are being divided into 8 different clusters using MiniBatchKMeans. The mean β values were between 0.1365 and 0.1973 and the number of maps in the clusters ranged between 25 and 176. See the boxplot in Fig. 11a. The spatial separation between each cluster is shown in the 2D PCA projection in Fig. 11b and the link between each cluster and the physical haze generation rules is provided in the strategy composition chart in Fig. 11c. For instance, the cluster 4 has a uniform haze and clusters 1 and 2 show heterogeneous strategies such as Perlin and fog-bank haze. This mapping illustrates the spatial representation of features in the network and how it links to different physical distributions of haze.

A detailed look at an individual test sample highlights this behavior. For instance, in one representative case, the predicted beta map achieves a remarkably low MAE of 0.0072. Yet, the corresponding direct physical inversion yields a poor reconstruction score of 9.77 dB PSNR and 0.5487 SSIM. Once the residual refinement is applied, the final joint output recovers to 22.30 dB PSNR and 0.8836 SSIM. This specific example demonstrates that minimal error in the estimated attenuation map does not guarantee high-quality physical dehazing. Because depth scale alignment, global airlight estimation, and the division by small transmission denominators are coupled, small parameter errors propagate and amplify

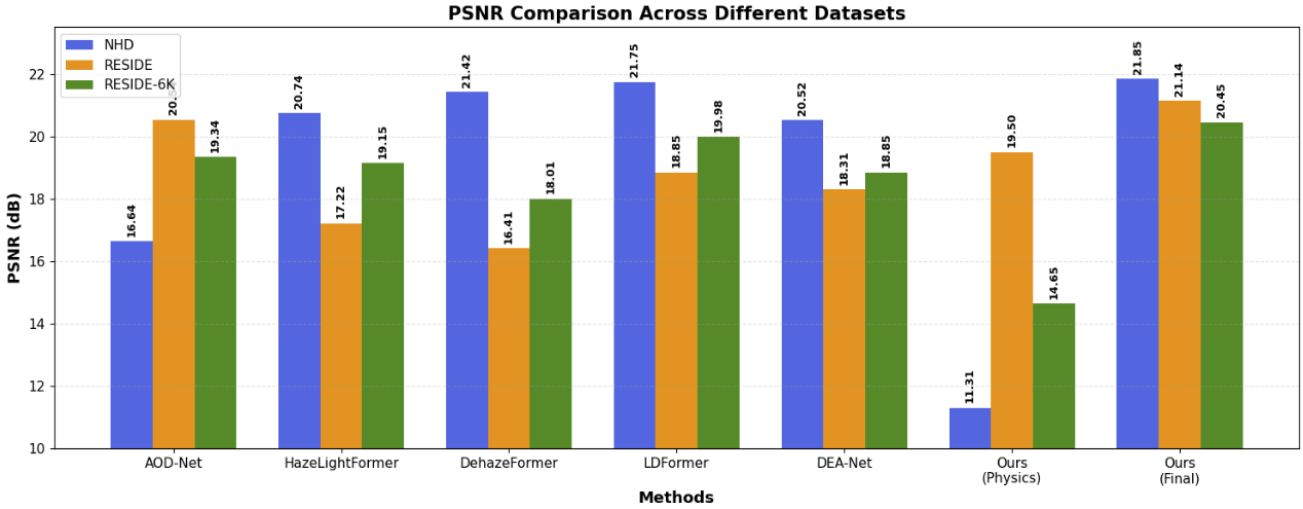
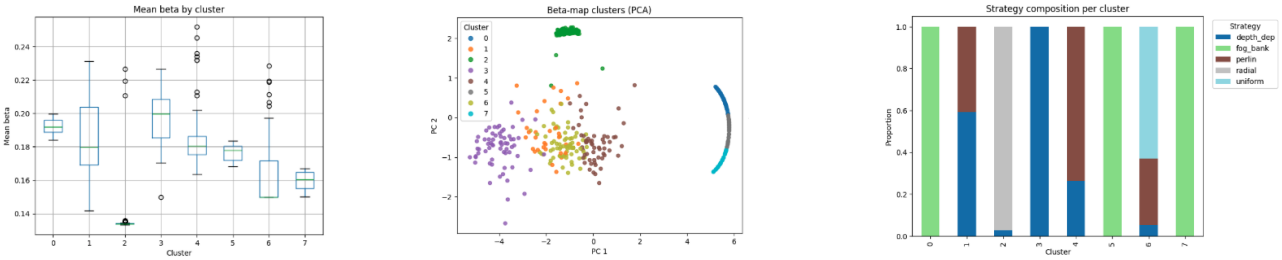


Fig. 10: Quantitative PSNR comparison of different dehazing methods across the NHD, RESIDE, and reside-6k datasets.



(a) Mean beta by cluster (b) Beta-map clusters (PCA) (c) Strategy composition per cluster

Fig. 11: Beta-map cluster analysis diagnostics: (a) boxplot of mean beta values, (b) 2D PCA projection of clustered beta maps, and (c) distribution of haze generation strategies within each cluster.

noise. Consequently, evaluating dehazing systems solely on parameter accuracy (like beta MAE) can be misleading; visual inspection of the error maps and transmission fields is essential to diagnose failure modes.

VI. Ablation Study

No ablation table is present in the supplied notebook, so the following is an ablation protocol rather than a reported result. The first experiment should compare direct beta decoding with the proposed retrieval-plus-refinement path. It should report beta MAE, beta RMSE, Pearson correlation, gradient correlation, image PSNR, SSIM, MAE, RMSE, parameter count, and runtime. A positive result would require performance gains with confidence intervals across at least three seeds, not only a single best checkpoint.

The second experiment should isolate the physical branch: compare final restoration from the physical inversion with a residual decoder fed the original hazy image, a decoder fed J_{phys}, and a direct end-to-end dehazer. This reveals whether beta/A/depth supervision improves restoration beyond adding network capacity. The third experiment should replace estimated depth with NHD ground-truth depth. The difference quantifies the error introduced by the monocular-depth bridge and separates beta estimation error from depth-scale error.

Further ablations should remove L_{edge}, L_{smooth}, L_A, L_{rec}, L_{phys}, SSIM, and LPIPS one at a time; compare beta grids of 14, 28, 56, and 112; and report results separately for uniform, fog-bank, Perlin, radial, and depth-dependent haze. The proposed DARTS decoder must also be evaluated against the hand-designed decoder with the same training budget. Without these controls, a final-model gain cannot be assigned to a particular component.

A reporting table for every ablation should contain the configuration name, NHD beta MAE/RMSE/Pearson/gradient correlation, NHD PSNR/SSIM/MAE/RMSE, external PSNR/SSIM, parameter count, FLOPs, and inference latency. The baseline direct-beta model should be retrained for each seed, rather than comparing one seed of the extension with a different seed of the baseline. Where external data lack beta labels, the table should show a dash rather than impute a value.

VII. Computational Complexity

The evaluated model contains approximately 23.98 M trainable parameters. Its main computational costs are the ViT-S en-

coder, beta upsampling decoder, separate Depth Anything V2 inference, and residual encoder-decoder. The 56×56 beta grid is a practical compromise: it preserves spatial variation while avoiding the memory cost of full-resolution beta prediction. Transmission is bilinearly upsampled before physical inversion. The reported model does not perform codebook retrieval or DARTS search at inference; their cost must therefore not be included in reported runtime.

For a future retrieval model, codebook storage is $O(KHW)$ for K beta prototypes with spatial size $H \times W$, or $O(KD)$ if retrieving latent vectors. With $K = 512$ and $H = W = 56$, a float32 beta-map codebook uses roughly 6.1 MiB before auxiliary metadata. Dense similarity evaluation can be made $O(KD)$ per image with latent queries, while direct map retrieval is more expensive. DARTS raises search-time cost because candidate operations are jointly active; after discretization, deployment cost should be measured from the selected architecture only.

VIII. Limitations and Future Work

The dataset has strong physics labels but limited scene diversity: 555 training pairs arise from 112 clear images. Multiple haze variants improve atmosphere diversity but do not replace novel scene content. The system uses per-image normalized relative depth, so beta values are meaningful within the adopted depth convention rather than as independently calibrated optical coefficients. External results provide output-level transfer evidence, not beta-parameter validation.

Future work should restore and version reside-6k with checksums and explicit pairing rules; add established dehazing baselines under identical preprocessing; report confidence intervals; and include real-haze evaluation or no-reference perceptual measures [28]. The codebook and DARTS extensions should be implemented as separate controlled experiments, with their status transparently changed from proposed to evaluated only after reproducible results are available.

IX. Conclusion

The supplied implementation is a physics-guided hybrid dehazer that combines dense beta/A estimation, relative-depth-conditioned atmospheric inversion, and a residual restoration decoder. Its NHD clean-image split is valid and identity-disjoint, and recorded metrics show that the final decoder substantially improves the direct physics output on NHD and the recorded reside-6k subset. The external RESIDE result also demonstrates that correction is domain-sensitive. This detailed paper adds a complete, honest extension plan for codebook retrieval and DARTS search while keeping them separate from the evaluated direct-beta system. The next credible step is a multi-seed, baseline-matched ablation study with restored external data provenance.

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